Ensemble-Based Spatial Snow Water Equivalent Estimates Using the SWE Evaluation Tool for the Rio Grande Basin

March 15, 2026

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Introduction

No spaceborne sensor can directly measure snow water equivalent (SWE). Therefore, the spatial distribution of SWE must be estimated from remotely sensed observations combined with modeling approaches. Existing gridded SWE products exhibit documented accuracy limitations across both mountainous and non-mountainous regions, resulting in a wide range of SWE estimates for any given location. To address this uncertainty, the New Mexico Office of the State Engineer (NM OSE), in collaboration with the Institute of Arctic and Alpine Research (INSTAAR), developed the SWE Evaluation Tool (SWEET). SWEET integrates multiple suitable SWE data products into an ensemble framework, with the objective of producing near-real-time spatial SWE estimates for the Rio Grande Basin that are, on average, as accurate as or more accurate than any individual product alone.

Figure 1 shows the near-real-time (March 15, 2026) ensemble SWE estimates for the Rio Grande Basin. Subsequent sections of this report present detailed ensemble-based SWE maps, along with summaries by subbasin and elevation band. In addition to the ensemble results highlighted in this report, figures and tables for each individual SWE product are also provided for download to support transparency and further analysis. These data sets and visualizations are available on the GitHub repository.

For ease of navigation, hyperlinks are provided throughout this report to help readers quickly access basin-specific sections and supporting data tables.

Rio Grande Basin SWE Overview

Rio Grande Headwaters

Upper Rio Grande

Rio Grande-Elephant Butte (Middle Rio Grande)

About SWEET

Rio Grande Basin SWE Overview

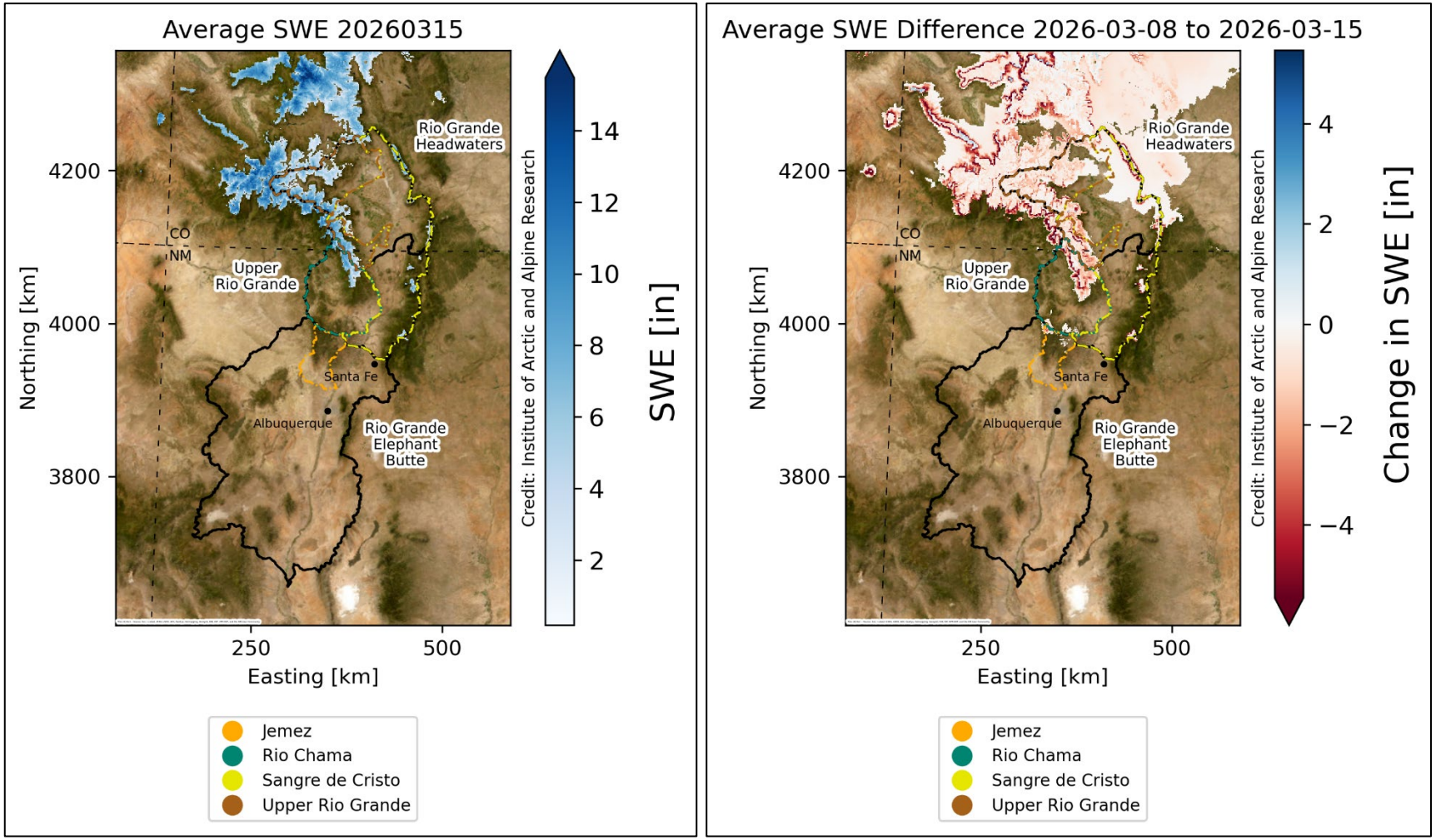

Figure 1. Estimated ensemble SWE (in.) and change in estimated ensemble SWE (in.) across the Rio Grande Basin for March 15, 2026. *SWE amounts (in.) across the Rio Grande Basin (left); dark blue pixels denote greater SWE amounts. Change in SWE amounts (in.) across the Rio Grande Basin from March 08, 2026, to March 15, 2026 (right); blue pixels denote an increase in SWE amounts whereas red pixels denote a decrease in SWE amounts.*

Table 1. Estimated ensemble SWE by HUC6 Basin for March 15, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since March 08, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value. Values are separated by HUC8 subbasin and totaled for each HUC8 basin (bold).*

HUC6 Basin	HUC8 Basin	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-03-08 (af)	Median Vol. Change since 2026-03-08 (af)
Rio Grande Headwaters	Rio Grande Headwaters	301,335	297,599	295,581 (UA_SWE)	310,824 (CU_SWE)	-37,522	-37,273
	Alamosa-Trinchera	79,038	86,988	57,021 (SNODAS)	93,106 (UA_SWE)	-24,435	-20,852
	San Luis	53,400	51,350	50,082 (SNODAS)	58,767 (UA_SWE)	-15,287	-15,762
	Saguache	51,544	50,416	43,861 (SNODAS)	60,355 (UA_SWE)	-17,746	-14,855
	Conejos	120,142	116,176	112,165 (CU_SWE)	132,085 (SNODAS)	-25,639	-28,287
	Total	**605,459**	**611,743**	**580,648 (SNODAS)**	**623,985 (UA_SWE)**	**-120,629**	**-99,995**
Upper Rio Grande	Upper Rio Grande	55,160	61,634	35,506 (CU_SWE)	68,341 (SNODAS)	-17,464	-12,723
	Rio Chama	109,178	104,463	79,599 (UA_SWE)	143,471 (SNODAS)	-35,421	-35,867
	Total	**164,338**	**141,233**	**139,969 (CU_SWE)**	**211,812 (SNODAS)**	**-52,885**	**-46,713**
Rio Grande-Elephant Butte	Rio Grande-Santa Fe	554	237	164 (CU_SWE)	1,261 (SNODAS)	-187	-152
	Jemez	1,544	1,305	361 (UA_SWE)	2,965 (CU_SWE)	-2,821	-3,035
	Rio Grande-Albuquerque	0	0	0	0	-8	0
	Arroyo Chico	0	0	0	0	0	0
	North Plains	0	0	0	0	0	0
	Rio San Jose	0	0	0	0	0	0
	Plains of San Agustin	0	0	0	0	0	0
	Rio Salado	0	0	0	0	0	0
	Jornada Del Muerto	0	0	0	0	0	0
	Elephant Butte Reservoir	0	0	0	0	0	0
	Rio Puerco	610	268	93 (UA_SWE)	1,468 (CU_SWE)	-745	-815
	Total	**2,707**	**2,834**	**691 (UA_SWE)**	**4,597 (CU_SWE)**	**-3,761**	**-4,030**

Table 2. Estimated ensemble SWE by specified subbasin groupings for March 15, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since March 08, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value. Values are separated by HUC8 subbasin and totaled for each specified subbasin grouping (bold).*

OSE Grouping	HUC8 Basin	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-03-08 (af)	Median Vol. Change since 2026-03-08 (af)
Upper Rio Grande	Rio Grande Headwaters	301,335	297,599	295,581 (UA_SWE)	310,824 (CU_SWE)	-37,522	-37,273
	Saguache	51,544	50,416	43,861 (SNODAS)	60,355 (UA_SWE)	-17,746	-14,855
	Conejos	120,142	116,176	112,165 (CU_SWE)	132,085 (SNODAS)	-25,639	-28,287
	Total	**473,021**	**473,405**	**472,112 (UA_SWE)**	**473,546 (SNODAS)**	**-80,907**	**-84,722**
Sangre de Cristo	San Luis	53,400	51,350	50,082 (SNODAS)	58,767 (UA_SWE)	-15,287	-15,762
	Alamosa-Trinchera	79,038	86,988	57,021 (SNODAS)	93,106 (UA_SWE)	-24,435	-20,852
	Upper Rio Grande	55,160	61,634	35,506 (CU_SWE)	68,341 (SNODAS)	-17,464	-12,723
	Total	**187,598**	**175,444**	**173,845 (CU_SWE)**	**213,506 (UA_SWE)**	**-57,186**	**-74,068**
Rio Chama	**Total**	**109,178**	**104,463**	**79,599 (UA_SWE)**	**143,471 (SNODAS)**	**-35,421**	**-35,867**
Jemez	**Total**	**1,544**	**1,305**	**361 (UA_SWE)**	**2,965 (CU_SWE)**	**-2,821**	**-3,035**

Upper Rio Grande Snowpack Update Application

The OSE/ISC Upper Rio Grande Snowpack Update Application (Application) provides water managers and users with detailed SNOTEL-based SWE data organized by the same subbasin groupings shown in Table 2. The Application is updated regularly from December through May and compiles key SWE information from SNOTEL sites within each subbasin. In addition to the Upper Rio Grande groupings (Upper Rio Grande, Sangre de Cristo, Rio Chama, and Jemez), the Application also includes the San Juan River Basin, with data summarized for the Upper San Juan, Animas, and Piedra subbasins.

The Application can be viewed here[1].

[1] Please note that access to the Application may be restricted to employees of the NM OSE/ISC at this time. We are currently working on public access.

Rio Grande Headwaters

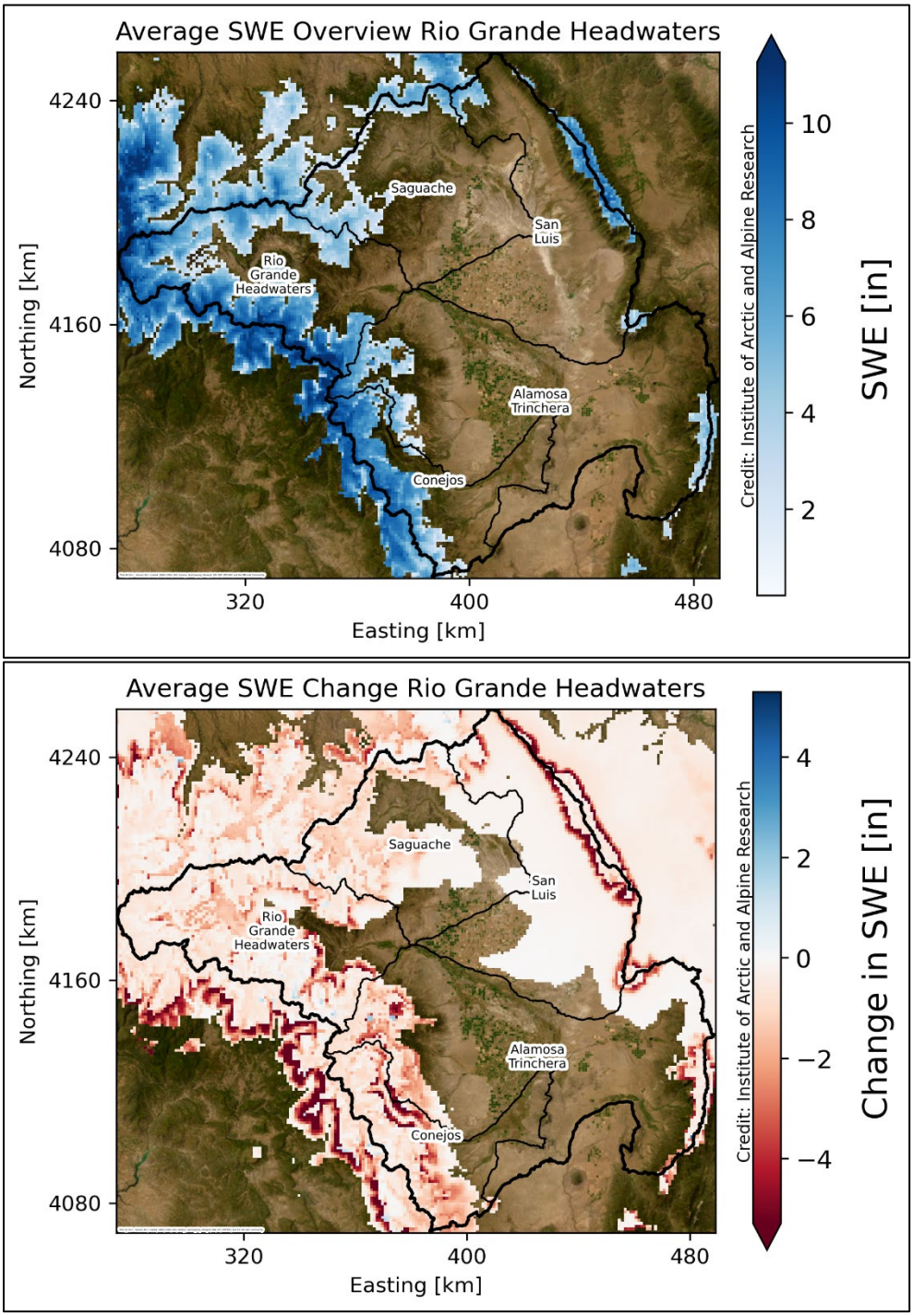

Figure 2. Estimated ensemble SWE (in.) and change in estimated ensemble SWE (in.) across the Rio Grande Headwaters HUC6 Basin for March 15, 2026, derived from the SWE Evaluation Tool (SWEET). *Ensemble SWE amount across the Rio Grande Headwaters Basin, showing spatial patterns of SWE across subbasins including Rio Grande Headwaters, Alamosa-Trinchera, San Luis, Saguache, and Conejos (top); dark blue pixels denote greater SWE amounts. Change in SWE amounts (in.) across the Rio Grande Headwaters from March 08, 2026, to March 15, 2026 (bottom); blue pixels denote an increase in SWE amounts whereas red pixels denote a decrease in SWE amounts. Basin and subbasin boundaries are shown in black.*

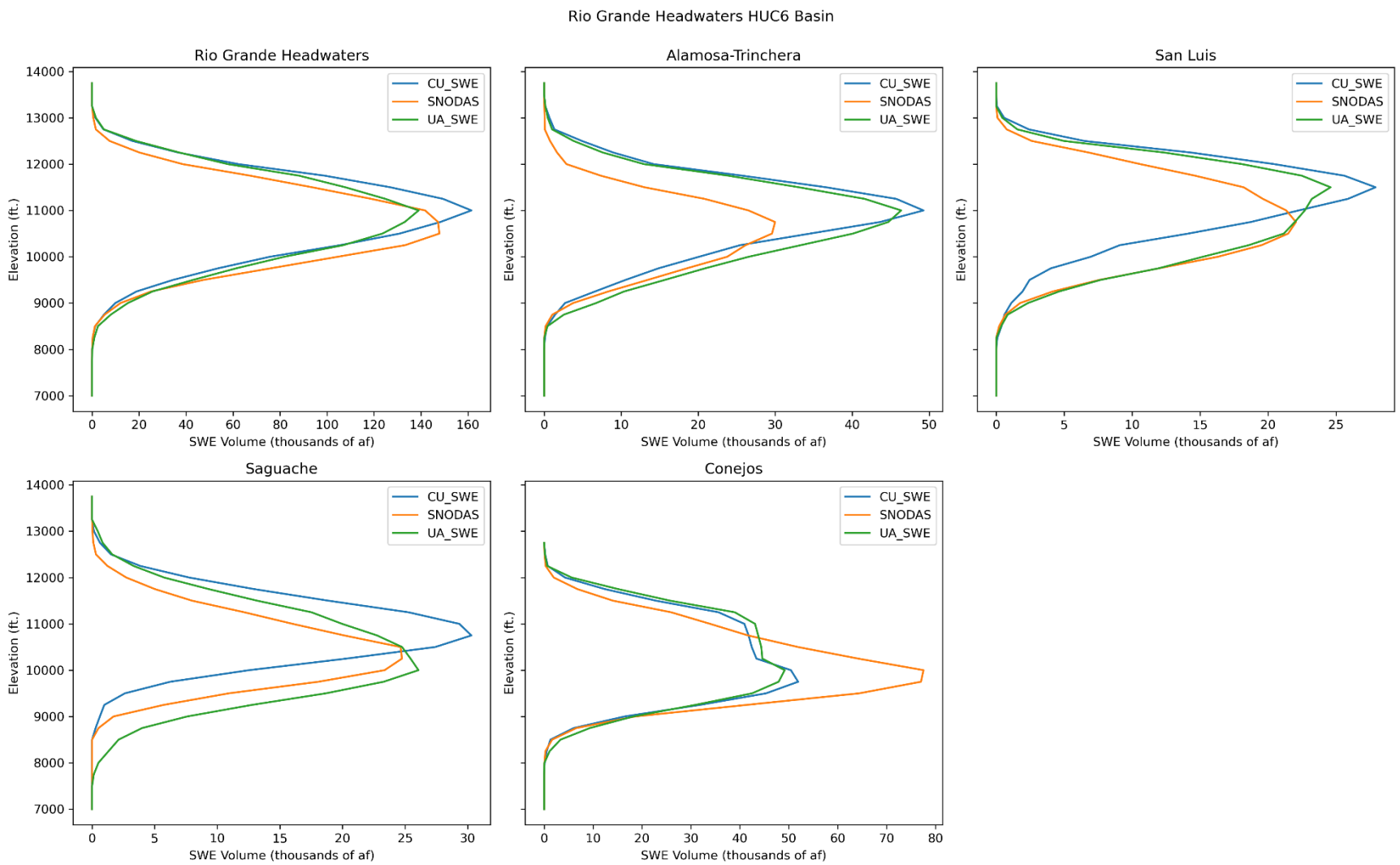

Figure 3. Estimated SWE volume (thousands of af) by subbasin and elevation band across the Rio Grande Headwaters HUC6 Basin for March 15, 2026. *Elevation-distributed SWE volume for the Rio Grande Headwaters HUC6 Basin and associated subbasins (Rio Grande Headwaters, Alamosa–Trinchera, San Luis, Saguache, and Conejos), comparing three spatial SWE products (CU_SWE, SNODAS, and UA_SWE). Curves show SWE volume summarized by elevation band, illustrating differences in both magnitude and vertical distribution among products across each subbasin.*

Table 3. Estimated ensemble SWE by HUC8 subbasin and elevation band across the Rio Grande Headwaters HUC6 Basin for March 15, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since March 08, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value.*

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-03-08 (af)	Median Vol. Change since 2026-03-08 (af)
Rio Grande Headwaters	Rio Grande Headwaters	7000-8000'	0	0	0	0	0	0
		8000-9000'	32	3	0 (CU_SWE)	93 (UA_SWE)	-749	-886
		9000-10000'	12,402	12,101	9,882 (CU_SWE)	15,224 (UA_SWE)	-8,552	-10,476
		10000-11000'	87,406	81,848	75,731 (CU_SWE)	104,639 (SNODAS)	-12,723	-11,370
		11000-12000'	147,415	141,769	139,002 (UA_SWE)	161,474 (CU_SWE)	-11,211	-7,318
		12000-13000'	52,893	57,779	38,566 (SNODAS)	62,335 (CU_SWE)	-4,186	306
		13000-14000'	1,187	1,404	521 (SNODAS)	1,635 (UA_SWE)	-100	-229
	Alamosa-Trinchera	7000-8000'	0	0	0	0	-327	-128
		8000-9000'	0	0	0	0	-546	-642
		9000-10000'	4,400	3,750	2,707 (CU_SWE)	6,743 (UA_SWE)	-8,533	-8,616
		10000-11000'	23,485	23,773	20,113 (CU_SWE)	26,570 (UA_SWE)	-7,743	-8,479
		11000-12000'	40,716	46,359	26,533 (SNODAS)	49,255 (CU_SWE)	-5,983	-1,542
		12000-13000'	10,027	12,965	2,897 (SNODAS)	14,220 (CU_SWE)	-1,247	-89
		13000-14000'	410	468	68 (SNODAS)	694 (CU_SWE)	-56	-1
	San Luis	7000-8000'	0	0	0	0	-3,112	-3,790
		8000-9000'	0	0	0	0	-1,953	-1,971
		9000-10000'	1,731	1,758	1,105 (CU_SWE)	2,330 (UA_SWE)	-2,938	-3,090
		10000-11000'	12,823	15,207	6,958 (CU_SWE)	16,304 (SNODAS)	-3,177	-3,052
		11000-12000'	22,093	22,247	21,334 (SNODAS)	22,697 (UA_SWE)	-2,239	-1,704
		12000-13000'	16,380	18,087	10,596 (SNODAS)	20,459 (CU_SWE)	-1,825	-56
		13000-14000'	373	447	90 (SNODAS)	581 (CU_SWE)	-43	1
	Saguache	7000-8000'	0	0	0	0	-1,129	-760
		8000-9000'	169	0	0 (SNODAS, CU_SWE)	507 (UA_SWE)	-1,364	-859
		9000-10000'	3,308	1,711	623 (CU_SWE)	7,591 (UA_SWE)	-5,140	-6,785
		10000-11000'	20,662	23,366	12,550 (CU_SWE)	26,071 (UA_SWE)	-7,522	-7,736
		11000-12000'	21,776	19,974	16,022 (SNODAS)	29,333 (CU_SWE)	-2,026	-807
		12000-13000'	5,416	5,757	2,734 (SNODAS)	7,757 (CU_SWE)	-554	-30
		13000-14000'	212	152	28 (SNODAS)	456 (UA_SWE)	-12	-24
	Conejos	7000-8000'	0	0	0	0	0	0

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-03-08 (af)	Median Vol. Change since 2026-03-08 (af)
Rio Grande Headwaters	Conejos	8000-9000'	78	66	57 (UA_SWE)	111 (CU_SWE)	-998	-1,155
		9000-10000'	17,739	18,218	16,420 (CU_SWE)	18,578 (SNODAS)	-13,067	-10,437
		10000-11000'	59,051	50,442	49,166 (UA_SWE)	77,545 (SNODAS)	-6,752	-10,549
		11000-12000'	39,323	40,931	33,929 (SNODAS)	43,110 (UA_SWE)	-4,395	-2,852
		12000-13000'	3,951	4,260	1,968 (SNODAS)	5,625 (UA_SWE)	-426	-882

Upper Rio Grande

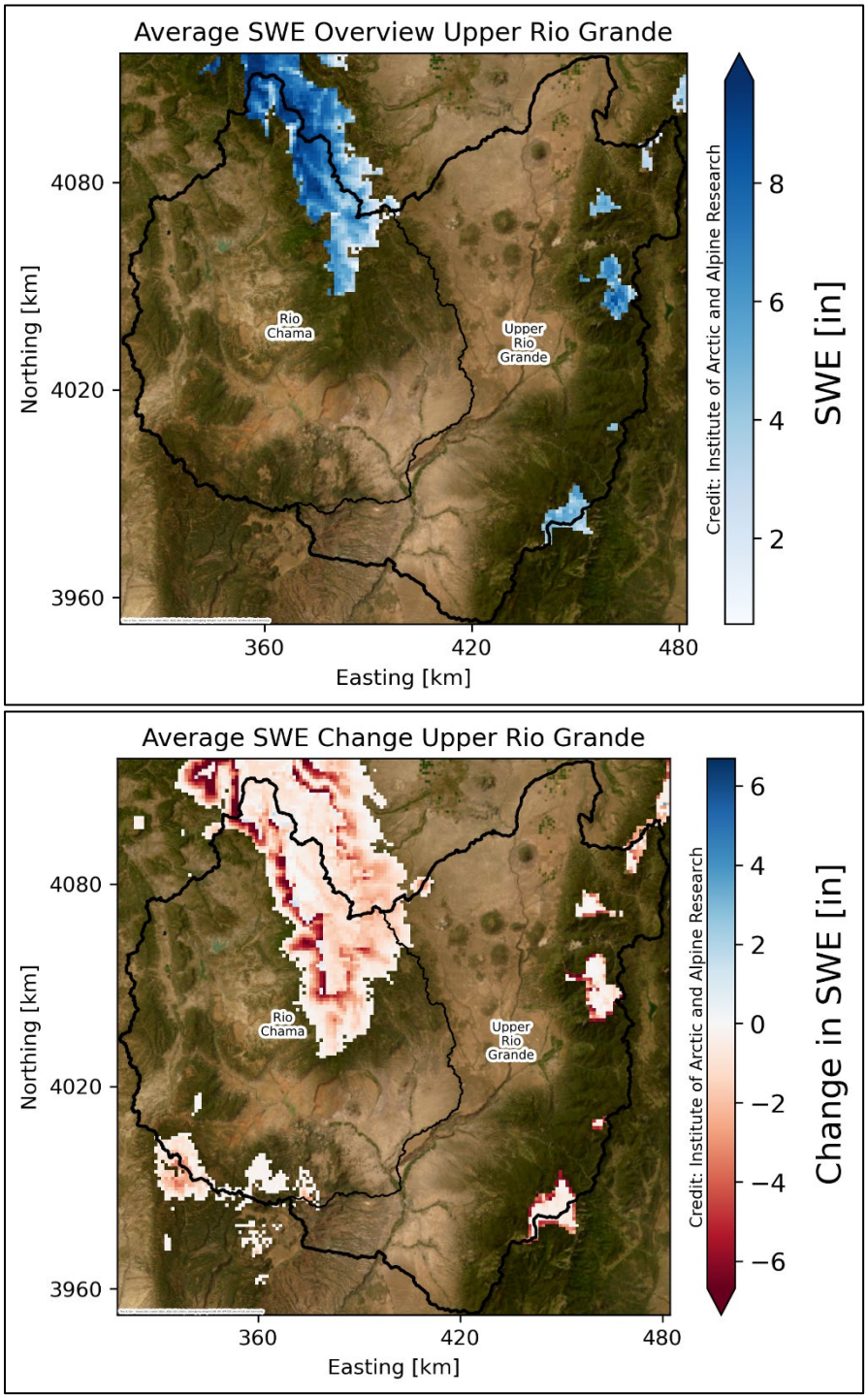

Figure 4. Estimated ensemble SWE (in.) and change in estimated ensemble SWE (in.) across the Upper Rio Grande HUC6 Basin for March 15, 2026, derived from the SWE Evaluation Tool (SWEET). *Ensemble SWE amount across the Upper Rio Grande Basin, showing spatial patterns of SWE across subbasins including Rio Chama and Upper Rio Grande (top); dark blue pixels denote greater SWE amounts. Change in SWE amounts (in.) across the Upper Rio Grande Basin from March 08, 2026, to March 15, 2026 (bottom); blue pixels denote an increase in SWE amounts whereas red pixels denote a decrease in SWE amounts. Basin and subbasin boundaries are shown in black.*

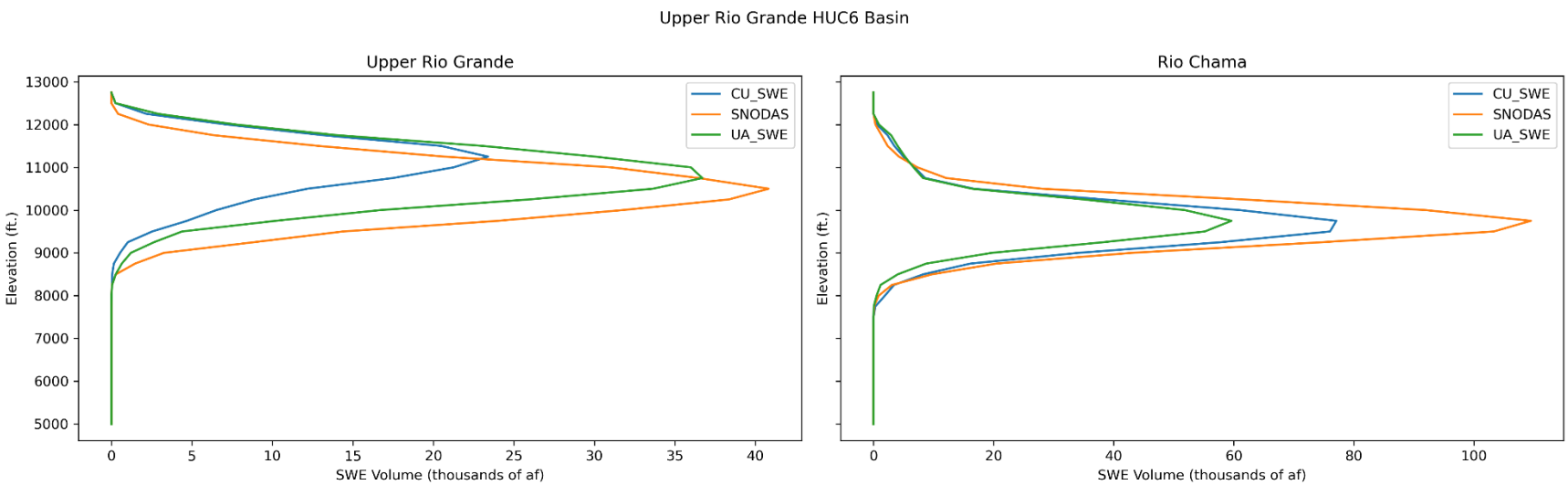

Figure 5. Estimated SWE volume (thousands of af) by subbasin and elevation band across the Upper Rio Grande HUC6 Basin for March 15, 2026. *Elevation-distributed SWE volume for the Upper Rio Grande HUC6 Basin and associated subbasins (Rio Chama and Upper Rio Grande), comparing three spatial SWE products (CU_SWE, SNODAS, and UA_SWE). Curves show SWE volume summarized by elevation band, illustrating differences in both magnitude and vertical distribution among products across each subbasin.*

Table 4. Estimated ensemble SWE by HUC8 subbasin and elevation band across the Upper Rio Grande HUC6 Basin for March 15, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since March 08, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value.*

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-03-08 (af)	Median Vol. Change since 2026-03-08 (af)
Upper Rio Grande	Upper Rio Grande	5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	-31	0
		8000-9000'	12	15	5 (UA_SWE)	17 (CU_SWE)	-357	-427
		9000-10000'	1,666	1,206	540 (CU_SWE)	3,251 (SNODAS)	-4,318	-2,198
		10000-11000'	18,309	16,682	6,515 (CU_SWE)	31,728 (SNODAS)	-7,993	-7,377
		11000-12000'	29,425	31,034	21,256 (CU_SWE)	35,986 (UA_SWE)	-3,939	-3,942
		12000-13000'	5,748	7,178	2,312 (SNODAS)	7,754 (UA_SWE)	-826	-649
	Rio Chama	5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	-273	-9
		8000-9000'	1,084	856	505 (UA_SWE)	1,892 (CU_SWE)	-4,457	-3,460
		9000-10000'	32,219	34,070	19,646 (UA_SWE)	42,940 (SNODAS)	-21,491	-19,648
		10000-11000'	68,332	61,074	51,877 (UA_SWE)	92,046 (SNODAS)	-8,541	-12,402
		11000-12000'	6,945	6,865	6,660 (UA_SWE)	7,310 (SNODAS)	-606	-866
		12000-13000'	597	562	319 (SNODAS)	910 (UA_SWE)	-53	-123

Rio Grande-Elephant Butte (Middle Rio Grande)

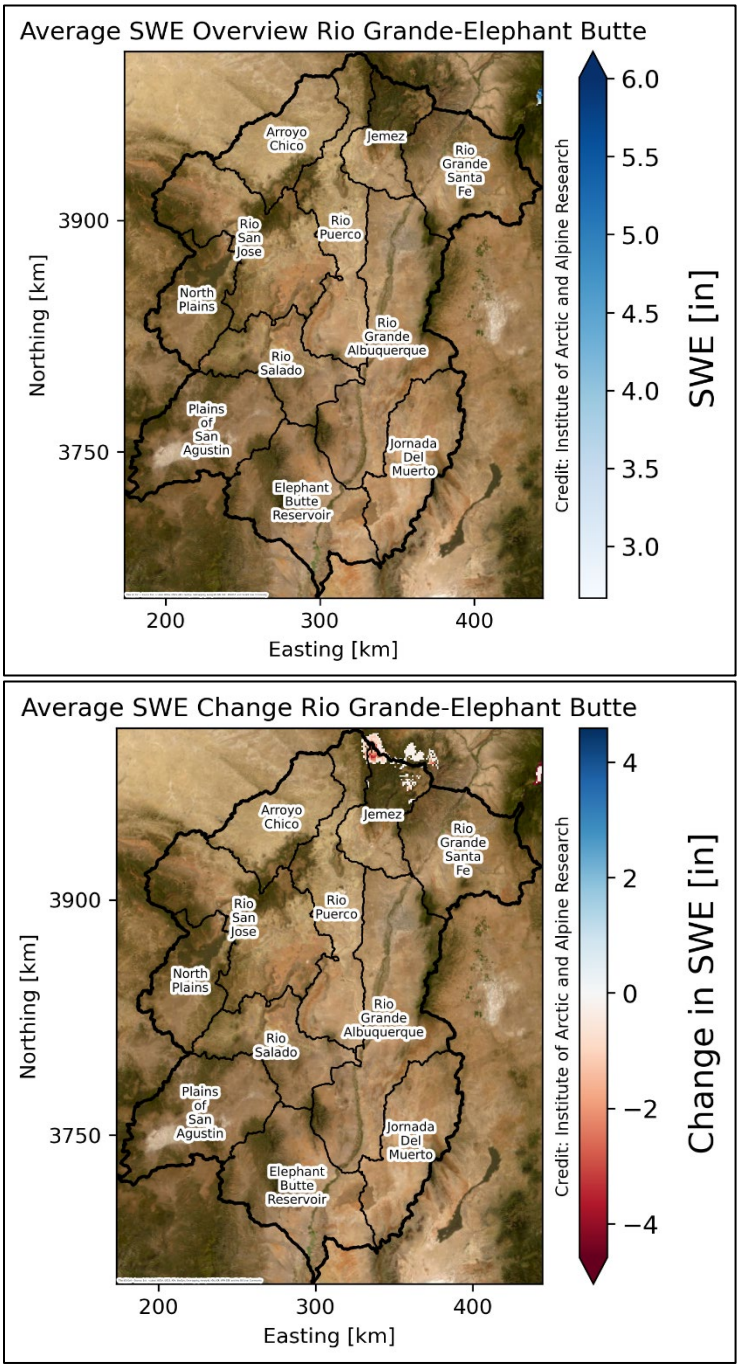

Figure 6. Estimated ensemble SWE (in.) and change in estimated ensemble SWE (in.) across the Rio Grande-Elephant Butte HUC6 Basin for March 15, 2026, derived from the SWE Evaluation Tool (SWEET). *Ensemble SWE amount across the Rio Grande-Elephant Butte Basin, showing spatial patterns of SWE across subbasins including Rio Grande-Santa Fe, Jemez, Arroyo Chico, Rio San Jose, Rio Puerco, Rio Grande-Albuquerque, Rio Salado, North Plains, Plains of San Agustin, Elephant Butte Reservoir, and Jornada Del Muerto (top); dark blue pixels denote greater SWE amounts. Change in SWE amounts (in.) across the Rio Grande-Elephant Butte Basin from March 08, 2026, to March 15, 2026 (bottom); blue pixels denote an increase in SWE amounts whereas red pixels denote a decrease in SWE amounts. Basin and subbasin boundaries are shown in black.*

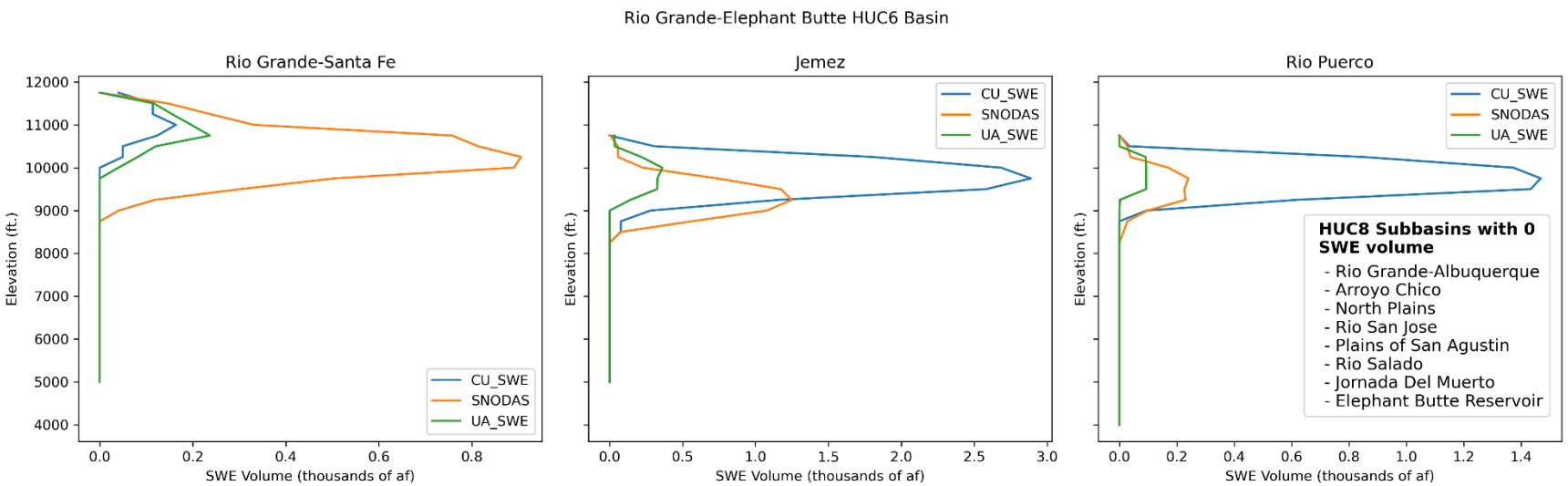

Figure 7. Estimated SWE volume (thousands of af) by subbasin and elevation band across the Rio Grande-Elephant Butte HUC6 Basin for March 15, 2026. *Elevation-distributed SWE volume for the Rio Grande-Elephant Butte HUC6 Basin and associated subbasins (Rio Grande-Santa Fe, Jemez, Arroyo Chico, Rio San Jose, Rio Puerco, Rio Grande-Albuquerque, Rio Salado, North Plains, Plains of San Agustin, Elephant Butte Reservoir, and Jornada Del Muerto), comparing three spatial SWE products (CU_SWE, SNODAS, and UA_SWE). Curves show SWE volume summarized by elevation band, illustrating differences in both magnitude and vertical distribution among products across each subbasin.*

Table 5. Estimated ensemble SWE by HUC8 subbasin and elevation band across the Rio Grande-Elephant Butte HUC6 Basin for March 15, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since March 08, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value.*

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-03-08 (af)	Median Vol. Change since 2026-03-08 (af)
Rio Grande-Elephant Butte	Rio Grande-Santa Fe	5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	-19	0
		9000-10000'	14	0	0 (UA_SWE, CU_SWE)	41 (SNODAS)	-53	-71
		10000-11000'	310	40	0 (CU_SWE)	890 (SNODAS)	-67	-28
		11000-12000'	230	197	164 (CU_SWE)	331 (SNODAS)	-47	-43
	Jemez	5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	-23	0
		8000-9000'	0	0	0	0	-354	-54
		9000-10000'	453	281	0 (UA_SWE)	1,077 (SNODAS)	-1,580	-1,391
		10000-11000'	1,091	361	228 (SNODAS)	2,684 (CU_SWE)	-863	-990
	Rio Grande-Albuquerque	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
		9000-10000'	0	0	0	0	-8	0
		10000-11000'	0	0	0	0	0	0
	Arroyo Chico	5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
		9000-10000'	0	0	0	0	0	0
		10000-11000'	0	0	0	0	0	0
	North Plains	6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
	Rio San Jose	5000-6000'	0	0	0	0	0	0

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-03-08 (af)	Median Vol. Change since 2026-03-08 (af)
Rio Grande-Elephant Butte	Rio San Jose	6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
		9000-10000'	0	0	0	0	0	0
		10000-11000'	0	0	0	0	0	0
	Plains of San Agustin	6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
		9000-10000'	0	0	0	0	0	0
	Rio Salado	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
		9000-10000'	0	0	0	0	0	0
	Jornada Del Muerto	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
	Elephant Butte Reservoir	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
		9000-10000'	0	0	0	0	0	0
		10000-11000'	0	0	0	0	0	0
	Rio Puerco	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	-41	0
		8000-9000'	0	0	0	0	-64	0
		9000-10000'	64	95	0 (UA_SWE)	97 (SNODAS)	-319	-207

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-03-08 (af)	Median Vol. Change since 2026-03-08 (af)
Rio Grande-Elephant Butte	Rio Puerco	10000-11000'	546	170	93 (UA_SWE)	1,374 (CU_SWE)	-321	-327

About SWEET

The following information provides a brief overview of SWEET, and the SWE products currently included in the ensemble framework. For more detailed descriptions, citations, and additional information, please refer to the GitHub repository.

SWE products currently used include:

1. **SNODAS SWE**
    a. The SNOw Data Assimilation System (SNODAS) is a physically based snow model developed by NOAA's National Operational Hydrologic Remote Sensing Center and operational since 2004. SNODAS produces daily 1-km gridded estimates of SWE and related snow variables by combining modeled weather inputs with satellite, airborne, and in situ snow observations.
2. **UA/SWANN SWE**
    a. The University of Arizona/Snow Water Artificial Neural Network Modeling System (UA/SWANN) is a research SWE product that uses snow models, artificial neural networks, and assimilated in situ measurements to generate daily gridded SWE and snow depth. The dataset provides 4-km and downscaled 800-m SWE across the conterminous United States, incorporating SNOTEL and COOP observations with PRISM precipitation and temperature fields.
3. **CU SWE**
    a. The University of Colorado (CU) SWE-Fusion product is an operational dataset produced by INSTAAR that provides near–real-time SWE estimates at 500-m resolution. The approach fuses SNOTEL observations with historical SWE reanalyses and terrain variables using a generalized linear model, and scales results using satellite-derived fractional snow-covered area from Snow Today.

The following notes describe key processing methods, assumptions, and visualization conventions used in generating SWE Evaluation Tool figures, tables, and elevation plots.

Figures:

- For ensemble figures, CU SWE and UA SWE are reprojected to the SNODAS resolution (1 sq km) and averages are only taken at pixels where at least 2 of the 3 products have SWE values greater than 0 inches.
- All overview and subbasin plots set pixels with values of 0 inches to NA so that the basemap is visible.
- For individual product figures, each product is displayed at its native resolution.
- Elevation bands of 250 feet are used in the elevation figures, and SWE volume is calculated for each product's native resolution in each band.
- Difference plots display pixels where snow was present in either the current or previous report.
- Figures are available in PNG format on the GitHub repository.

Tables:

- For all tables, mean, median, and volume are calculated at each product's native resolution - zeros are not removed from mean and median calculations.
- For ensemble tables, mean and median volume are calculated using the total volumes from the 3 products - not from an average aggregate raster.
- For min. and max., if 2 products have 0 volume, both products are listed in that cell. If all 3 products have 0 volume, no product is listed for min. or max. in that row.
- Change in SWE will only be calculated starting with the second report of the water year.
- Ensemble calculations: For HUC8 subbasins, the ensemble (mean and median) SWE and volume change (mean and median) are calculated individually for each subbasin. For the HUC6 "Total" row (tables 1 and 2), the ensemble is recalculated using the full HUC6 basin extent rather than summing HUC8. While the ensemble mean is additive and therefore consistent between HUC8 sums and the HUC6 total, the median reflects the middle value among the three products for the entire basin. As a result, median values will not sum to the HUC6 total, since they represent the difference between basin-wide medians rather than the sum of subbasin medians.
- Tables are available in CSV and PDF format on the GitHub repository.
- Elevation tables presented in the report (Tables 3, 4, and 5) are combined into one table (Table 3) on the GitHub repository.

Acknowledgments

The work completed and presented in this report is funded by NASA Western Water Actions Office (WWAO), subcontract no. 1712706, as part of the *Rio Grande River Basin Needs Assessment Report*. The SWE Evaluation Tool is a collaborative effort from members at INSTAAR, University of Colorado Boulder, and the New Mexico Office of the State Engineer.

We would also like to extend our appreciation to Noah Molotch and the Mountain Hydrology Group at CU Boulder, INSTAAR, for granting our team access to the CU SWE operational SWE dataset.